Project title: EduGenie -Google Gemini Powered Learning Assistant

Phase 5: Project Development Phase

1. Project Development Phase:

-The project development phase is the main implementation stage of EduGenie. During this phase, the planned design and requirements were converted into a working application.

-The development environnmet was first configured using Visual Studio Code and Python. A separate project environment was created and the required Python packages were installed.

-The backend was developed using FastAPI. FastAPI provides the API structure required for receiving requests from the frontend and returning responses.

-Google Gemini was integrated into the backend to provide Generative AI capabilities. The Gemini API key was configured securely using environment variables instead of directly exposing it inside the source code.

-The application receives the student's input and sends it to the appropriate backend function. The backend communicates with the Gemini model and receives the generated response. The response is then returned to the frontend.

Different project modules were also developed to support various learning activities.

Main Learning Features:

Question and Answer:

Students can enter questions and receive AI-generated answers.

Explanation:

Difficult concepts can be explained in a simpler and understandable manner.

Quiz:

The application can support students by generating practice questions based on learning topics.

Summarization:

Long learning content can be converted into shorter and important points.

Learning Assistance:

EduGenie provides interactive AI-based support to help students during their learning process.

-The frontend was developed to provide a simple and interactive experience. The overall application was connected together and tested during development.